Repeated Games (Not in the Book)

ECON 5001
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Where We Left Off

Prisoners' Dilemma:

	Cooperate	Defect
Cooperate	2, 2	0, 3
Defect	3, 0	1, 1

Think of it as a price war (deck 03): Dos Hermanos vs. Pitabilities

- Cooperate = keep prices Normal. Defect = cut prices
- Same block, **every day**: a repeated game!

- Play once: D is strictly dominant
- Play repeatedly: Can we enforce Cooperation?

MobLab: “Prisoner’s Dilemma (Matrix)”

- MobLab time!
- Twice-repeated PD game
- Play the PD game twice with the same partner
- Repeat with a new partner
  - They can’t see how you played previously

Feel free to talk! And play it the way you’d actually play it.

Playing Twice Can Introduce New Nash Equilibria

Example

Player 1 \ Player 2	L	C	R
T	1, 1	5, 0	0, 0
M	0, 5	4, 4	0, 0
B	0, 0	0, 0	3, 3

Pure NE: _____

"Best" Outcome: _____

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Is (M, C) a NE? _____

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Now play it **twice**:

- Period 1: play the game. Everyone sees what happened
- Period 2: play it again
- Total payoff $=$ period 1 payoff $+$ period 2 payoff

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Can (M, C) happen in period 1???

A Strategy With a Threat

Player 1 \ Player 2	L	C	R
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M	0, 5	4, 4	0, 0
B	0, 0	0, 0	3, 3

Period 1: Play (M, C)

Period 2:

- If (M, C) happened: play (B, R)
- Otherwise: play (T, L)

Reward with the good NE. Punish with the bad NE

Check P1 in period 1:

- Conform (M): $____ + ____ = ____$
- Deviate (T): $____ + ____ = ____$

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Check P1 in period 1:

- Conform (M): $____ + ____ = ____$
- Deviate (T): $____ + ____ = ____$

Period 2? Playing a NE either way, no incentive to deviate

Every Period-2 Subgame

After period 1, there are _____ possible **histories**.

History: What's happened so far in the game ((T, L), (T, C), (M, B), etc.)

Period 2: Strategies can respond to the history

After history (M, C): play (B, R)
After any other history: play (T, L)

SPNE: Solve via backward induction.

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Lesson: Period 2 can be used to **punish** & **reward** period 1 behavior

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Lesson: Period 2 can be used to **punish** & **reward** period 1 behavior
BUT... what happens in Period 2 better be a NE. Otherwise, not credible

Formalizing Repeated Games

Stage Game vs. Supergame

- **Stage game** G: the game played each period
  - Our 3×3 game, or the PD
- **Supergame** G^T: play G in periods $t = 1, 2, \ldots, T$
  - Everyone sees all past play
  - T can be finite or ∞

Action: what you do in **one** period

- $a_i^t \in A_i$ (eg, M)
- Profile: $a^t = (a_1^t, a_2^t)$

Strategy: a complete plan for G^T

- What to do in **every** period
- ...after **everything** that could happen

Histories

History at the start of period t: everything that happened before t

$$h_t = (a^1, a^2, \ldots, a^{t-1})$$

- $h_1 = \emptyset$ (nothing has happened yet)
- $H_t =$ set of all possible histories at the start of period t

Ex: Suppose the 3×3 game is played 4 times:

- $h_2 = (M, C)$ is one history at start of period 2
- $h_3 = ((M, C), (T, L))$ is a possible history at start of period 3
- $h_4 = ((M, C), (T, L), (M, C))$ is a possible period-4 history
- $|H_2| =$ _____ $|H_3| =$ _____ $|H_4| =$ _____

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Each history starts a **subgame** (it's a singleton node in the tree)

Pure Strategy in a Repeated Game

Definition
A **pure strategy** for Player i is a function s_i that maps every history to an action: $s_i(h_t) \in A_i$ for every h_t in every H_t

Back to our 2-period 3×3 game. Our threat strategy for P1:

$$
\begin{aligned}
s_1(\emptyset) &= _____ \\
s_1\big((M, C)\big) &= _____ && \text{when } h_2 = (M, C) \\
s_1(h_2) &= _____ && \text{for every other } h_2 \neq (M, C)
\end{aligned}
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P1 moves after how many different histories? $1 + ____ = ____$
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Imagine chess...

Two Kinds of Equilibrium

Stage-game NE: a NE of G (one period)

- (T, L) and (B, R) in our example

Equilibrium of the supergame: a NE or SPNE of G^T

- A profile of **strategies** (s_1, s_2)**,** not actions
- SPNE: a NE in the subgame after **every** history

Ex: our threat strategies are a SPNE of G^2

- Period-1 play of (M, C) is part of a SPNE
- Even though (M, C) is **not** a stage-game NE!

Stage NE Every Period Is Always Safe

Suppose $a^* = (a_1^*, \ldots, a_n^*)$ is a NE of the stage game G. ex1 (T, L)

Consider the strategy: $s_i(h_t) = a_i^*$ regardless of h_t
"Always play a_i^*, **regardless** of what happened in the past."

Theorem
If $a^* = (a_1^*, \ldots, a_n^*)$ is a NE of the stage game G, then all players playing the strategy "$s_i(h_t) = a_i^*$ for every h_t" is a SPNE of G^T.

Same for any **sequence** of stage NE that ignores history: $a^{1*}, a^{2*}, \ldots$

Why?

- Deviating today can't improve today's payoff (b/c it's a NE)
- Deviating today can't change tomorrow's payoff (b/c they'll still play a^*)
- Thus, deviating never improves payoffs

Corollary: Stage game G is a finite game $\Rightarrow$ G has a NE $\Rightarrow$ There exists an SPNE of G^T

Twice-Repeated Prisoners' Dilemma

Two-Period PD

	Cooperate	Defect
Cooperate	2, 2	0, 3
Defect	3, 0	1, 1

Price war over two days

Try a punishment strategy:

- Period 1: (C, C)
- Period 2: reward if both Cooperated, punish otherwise

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	Cooperate	Defect
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Price war over two days

Try a punishment strategy:

- Period 1: (C, C)
- Period 2: reward if both Cooperated, punish otherwise

Problem: Period 2 plan isn't stage game NE

SPNE requires NE of stage game in final period

Period 2: $s^*(h_2) = (D, D)$ regardless of h_2

Two-Period PD

	Cooperate	Defect
Cooperate	2, 2	0, 3
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Price war over two days

Try a punishment strategy:

- Period 1: (C, C)
- Period 2: reward if both Cooperated, punish otherwise

Problem: Period 2 plan isn't stage game NE

Problem: only **one** stage-game NE: (D, D). No credible punishment/reward.

Period 1: no reason to C b/c Period 2 will always be (D,D)

Only SPNE: $s^*(h) = (D,D)$ at every history h even $h = \emptyset$

Any Finite Number of Periods

- Period T: last period. Play (D, D)
- Period $T-1$: can't affect period T play. Play (D, D)
- Period $T-2$: ... (D, D)
- ...
- Period 1: (D, D)

Any Finite Number of Periods

- Period T: last period. Play _____
- Period $T-1$: can't affect period T play. Play _____
- Period $T-2$: ...
- ...
- Period 1: _____

repeated game

Theorem
If G has a unique NE and T is finite, the only SPNE of G^T is to play that NE every period.

e.g. Prisoners Dilemma

Any Finite Number of Periods

- Period T: last period. Play _____
- Period $T-1$: can't affect period T play. Play _____
- Period $T-2$: ...
- ...
- Period 1: _____

Theorem
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The game **unravels** from the end.

But In the Lab...

Lab result: people **do** cooperate in finitely repeated PDs

- Lots of cooperation early
- Collapses in the last few rounds ("end-game effect")
- With experience, the collapse starts earlier

More cooperation when players aren't sure the other is purely selfish

- A little doubt about the other's **type**...
- We'll see this in Chapter 15!

Source: Caplan, *A Practicum in Behavioral Economics*, §3, p. 176.
Selten & Stoecker (1986, *JEBO*); Andreoni & Miller (1993, *EJ*).

Infinitely Repeated Games: $T = \infty$

No Last Period

What if the game doesn't have a known end date? So, $T = \infty$

- Another interpretation: the game ends at a random time

Problem 1: Can't apply backward induction/SPNE

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Problem 1: Can't apply backward induction/SPNE

Problem 2: payoffs add up to infinity

- Always Cooperate: $2 + 2 + 2 + \cdots = \infty$
- Defect once, then get Defected on forever: $3 + 1 + 1 + \cdots = \infty$

Which would you prefer?
Can't compare ∞ to ∞

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Fix: future payoffs count **less** than today's

Discounting: Money Now vs. Later

Interest rate $r = 10\%$ (for example):

- At 10% interest, \$90.91 today grows to $\$90.91 \times 1.10 = \$100$ next year
- So \$100 next year is worth _____ today
- \$82.64 today grows to $\$82.64 \times 1.10 = \$90.91$ next year, then to \$100 in 2 years
- So \$100 in two years is worth _____ today

General: What amount x today will grow to \$100 in t periods?

$$x(1+r)^t = 100$$

$$x = \left(\frac{1}{1+r}\right)^t 100$$

$$x = \delta^t 100$$

Discount factor δ: how much we discount future payoffs per period: $\delta = 1/(1+r)$

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Skip the r and just work with δ.

δ can be a personal preference

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Patient: don't discount much. δ near 1 → tomorrow money is almost as good as today

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Skip the r and just work with δ.
Patient: don't discount much. δ near 1
Impatient: discount a lot. δ near 0 ⇒ doesn't care about future very much

Utility of a Stream of Payoffs

"Stream" of payoffs: u_i^1 today, u_i^2 next period, u_i^3 after that, *etc.*

"Exponential" discounting: add them up, discounting each period by another δ.
Total value of the stream of payoffs:

$$U_i = u_i^1 + \delta u_i^2 + \delta^2 u_i^3 + \cdots = \sum_{t=1}^{\infty} \delta^{t-1} u_i^t$$

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Fact: if $0 \leq \delta < 1$, this is **finite**

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$\delta = 1/2$

$x \frac{1}{1-\frac{1}{2}} = x \frac{1}{\frac{1}{2}} = 2x$

Fact: if $0 \leq \delta < 1$, this is **finite**

$1 + \delta 1 + \delta^2 1 + \delta^3 1 + \cdots = 1 \frac{1}{1-\delta} < \infty$

1 every day forever $= \$1$ today.

Important Example: Suppose the stream pays x every period forever:

$$U_i = x + \delta x + \delta^2 x + \delta^3 x + \cdots \qquad \delta U_i = \delta x + \delta^2 x + \delta^3 x + \delta^4 x + \cdots$$

$$U_i - \delta U_i = (1-\delta) U_i = x \qquad \Rightarrow \qquad U_i = x \frac{1}{1-\delta}$$

$x \frac{1}{1-\delta}$

The Infinitely Repeated PD

	Cooperate	Defect
Cooperate	2, 2	0, 3
Defect	3, 0	1, 1

One SPNE we already know: $s(h) = (D,D)$ for all h

- Always Defect, no matter what
- Why? See prev. slide

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Powerful strategy: **Grim Trigger:**

- Period 1: play Cooperate
- Keep Cooperating as long as **everyone** always has
- Once anyone Defects: you Defect **forever**
- Price war: High prices until someone undercuts. Then low prices forever

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Period $t = 1$

$$s_i(\emptyset) = C$$

Periods $t > 1$

$$s_i(h_t) = \begin{cases} C & \text{if } h_t = (\underbrace{(C,C),(C,C),\ldots,(C,C)}_{t-1 \text{ periods}}) \\ D & \text{for any other } h_t \end{cases}$$

(any D anywhere)

Is Grim Trigger a SPNE?

For SPNE, need to check that there's no profitable deviation in **any** subgame (history)

1. Someone already Defected ($h_t \neq ((C,C),(C,C),\ldots,(C,C))$) we saw at least one D

- Both Defect forever: (D,D) is the stage game NE every period
- Theorem $\Rightarrow$ no profitable deviation ✓

2. Everyone always Cooperated ($h_t = ((C,C),(C,C),\ldots,(C,C))$)

- Conform: $2 + 2\delta + 2\delta^2 + \cdots =$ $2\frac{1}{1-\delta}$ (C,C) forever ((C,C) → 2)
- Deviate (Defect today, triggering the Grim Trigger response tomorrow):
  - Today: 3
  - Then punished forever starting tomorrow: $\delta 1 + \delta^2 1 + \delta^3 1 + \cdots$ (D,D) forever after t=1
  - Total payoff from this plan:

$$3 + \delta 1 + \delta^2 1 + \delta^3 1 + \cdots$$
$$2 + (1 + \delta 1 + \delta^2 1 + \delta^3 1 + \cdots) = 2 + (\text{-----})$$
$$2 + \tfrac{1}{1-\delta} 1$$

What do we prefer?

The Threshold

In words: cooperating forever must beat grabbing 3 today and getting punished forever

Price war: is one day of extra sales from a price cut worth a price war forever?

Stick with the plan if

$$\underbrace{2\frac{1}{1-\delta}}_{\text{cooperate forever}} \geq \underbrace{2+\frac{1}{1-\delta}1}_{\text{grab today, punished forever}}$$

Stick with the plan

deviate

Solve for δ:

$$2\frac{1}{1-\delta} \geq 2+\frac{1}{1-\delta}$$

$$2 \geq 2(1-\delta)+1$$

$$2 \geq 2-2\delta+1$$

$$2\delta \geq 1$$

$$\delta \geq \tfrac{1}{2}$$

Stick with plan as long as $\delta \geq \frac{1}{2}$

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Result: Grim Trigger is a SPNE $\iff \delta \geq$ ½ ← for both players. In other words, if they're patient enough

Other Strategies

- **Tit-for-Tat:** start with Cooperate, then copy what they did last period
  - Won Axelrod's computer tournaments (1980)
  - Nice, retaliatory, forgiving
- **Always Cooperate**, no matter what
  - Is playing "Cooperate at every history" a SPNE? _____
- **Always Defect**: always a SPNE

Not in the book. Axelrod, *The Evolution of Cooperation* (1984). Tit-for-Tat was submitted by Anatol Rapoport.

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Lesson: cooperation needs a credible **threat**

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This Is Everywhere

Repeated PDs, with cooperate / defect in each:

- **Cartels:** OPEC members: stick to quota / pump more
- **Price wars:** airlines on a route: hold fares / slash fares
- **Trade:** countries: keep tariffs low / raise tariffs
- **Roommates:** do your chores / let them pile up
- **Online sellers:** ship what you promised / ship junk
- **WWI trenches:** live and let live / non-stop combat

In each: the **future** keeps people honest today

Trenches: “live and let live,” Axelrod (1984). Many historians disagree.

But Those Are More Complicated

Real versions add:

- **Imperfect monitoring:** you can't directly see what your opponent did
  - e.g., you don't see their price cut, but you see a drop in your demand
- **Many players:** whom do you punish?
- **Entry:** new players who weren't part of the original agreement
- **Renegotiation:** after a price war, both want to "let bygones be bygones"

Core tension survives: today's gain vs. tomorrow's punishment.
And, is tomorrow's punishment even credible?

How Much Can Repetition Do?

Minmax Payoff in the PD

	Cooperate	Defect
Cooperate	2, 2	0, 3
Defect	3, 0	1, 1

- You're the row player
- Suppose your opponent wants to punish you for your past behavior

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In this game: Your minmax payoff is 1

Minmax Payoff

In words: the worst the others can hold you to, if you always best respond

$$v_i = \underbrace{\min_{a_{-i}}}_{\text{others hurt you}} \quad \underbrace{\max_{a_i} u_i(a_i, a_{-i})}_{\text{you best respond}}$$

Ex: in the previous slide, $v_i = 1$ for both players

Minmax Payoff

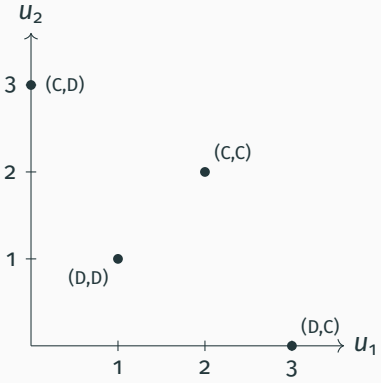
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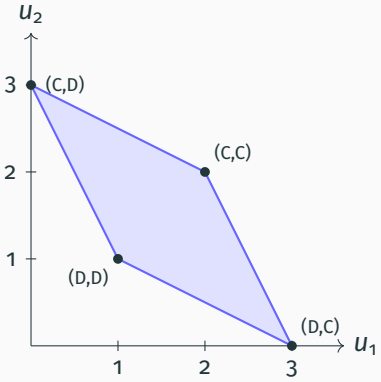
Fact: You can always get at least v_i. Thus, no SPNE can give you less than v_i

Feasible Payoffs



Per-period average payoffs

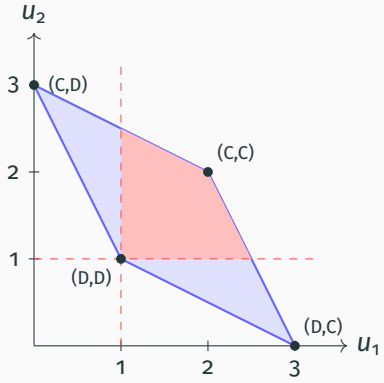
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Feasible: any mix of the four outcomes (mixing, taking turns, etc.)

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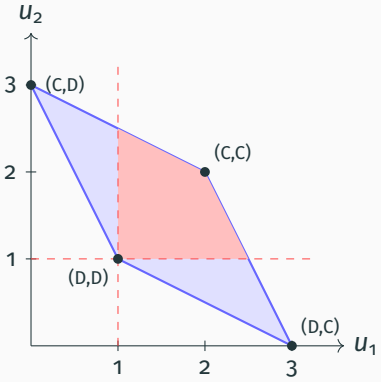


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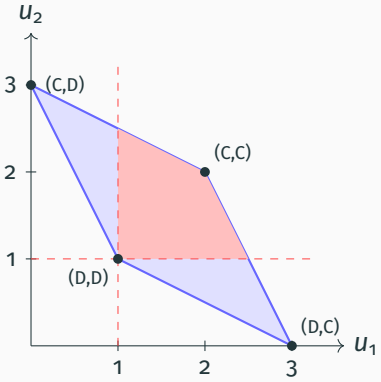
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SPNE #1: Grim Trigger gives payoffs $(2, 2)$

SPNE #2: Always Defect gives payoffs $(1, 1)$

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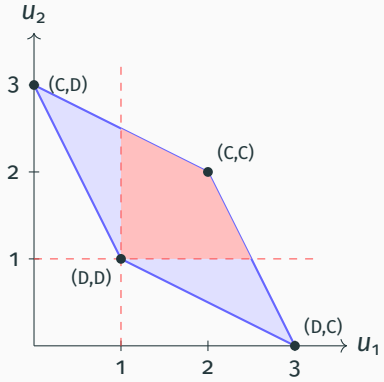
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The Folk Theorem

Folk Theorem (stated without proof)
In an infinitely repeated game, **any** feasible payoff giving every player more than their minmax v_i is the average payoff of some SPNE, if δ is close enough to 1.

- "Folk": everyone knew it before anyone published it
- PD: anything in the red region. Including (2,2)

Not in the book. Friedman (1971); Fudenberg & Maskin (1986, *Econometrica*). Some technical conditions apply.

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- **When** cooperation is possible: need $T = \infty$, patience ($\delta \to 1$), perfect monitoring
- **Comparative statics:** higher $\delta \Rightarrow$ more cooperation. Confirmed in the lab
- Explains **institutions**: reviews, credit scores, cartel meetings

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New question: which equilibrium? Focal points, talking, culture...

Let’s Work One Out

N. High & Hudson St.

Speedway and Sunoco both have stations at N. High & Hudson.

Every day each posts a **High** or **Low** price. Asymmetric PD: Speedway is bigger

Speedway \ Sunoco	High	Low
High	6, 4	1, 6
Low	8, 1	3, 2

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Consider Grim Trigger

Speedway:

Conform: $6 + \delta 6 + \delta^2 6 + \cdots = 6\frac{1}{1-\delta}$ Deviate: $8 + \delta 3 + \delta^2 3 + \cdots = 5 + 3\frac{1}{1-\delta}$

Grim Trigger is SPNE if $6\frac{1}{1-\delta} \geq 5 + 3\frac{1}{1-\delta} \Rightarrow \delta \geq 2/5 = 0.4$

Sunoco:

Conform: $4 + \delta 4 + \delta^2 4 + \cdots = 4\frac{1}{1-\delta}$ Deviate: $6 + \delta 2 + \delta^2 2 + \cdots = 4 + 2\frac{1}{1-\delta}$

Grim Trigger is SPNE if $4\frac{1}{1-\delta} \geq 4 + 2\frac{1}{1-\delta} \Rightarrow \delta \geq 1/2$

SPNE if Speedway has $\delta \geq 0.4$ and Sunoco has $\delta \geq 0.5$

What Have We Learned, Charlie Brown?

- **Stage game** vs. **supergame**. Actions vs. strategies
- A **history** is a list of all past actions
- Strategy: specifies an action for **every possible history**
- Stage game NE every period: always a SPNE
- Multiple stage NE: reward and punish, even with T finite
- Unique stage NE and T finite: **unraveling**
- $T = \infty$: discount with δ. Grim Trigger works if δ is big enough
- **Folk theorem**: any payoff above minmax is an SPNE payoff, if δ is close enough to 1

Reading & Practice

- Not in Bonanno. These slides are the reading
- Optional: Osborne, *An Introduction to Game Theory*, Chapters 14–15
- Optional: Caplan, §3, p. 176 (finitely repeated PD)
- Practice: redo the PD with $(C, C) = (2.5, 2.5)$. New threshold?